Mis-interpretation and misuse of p-values

Presentation Outline

Mis-interpretations of the p-value

Lots of mis-interpretations, and many reject any use of p-values

- The p-value:
  - *is not* the probability that the null hypothesis is true
  - *is not* the probability that the data were produced by random chance alone
  - *does not* measure the size of an effect
  - *does not* say anything about the importance of an effect
- A large p-value **does not** prove that the null hypothesis is true
  - When p large, e.g., 0.23, can claim "no evidence of a difference"
  - *can not* claim "there is no difference" (or "no effect")

Misuse of the p-value

Misuse of the p-value

- Researchers: Doing many tests, picking out those with $p < 0.05$
  - Tables of correlations between pairs of variables
  - e.g., 20 ways to quantify A, 15 ways to quantify B
  - 300 correlation coefficients, focus on the 3 that have $p < 0.05$
- Researchers: Looking for a relationship between A and B
  - Initial analysis has $p > 0.10$
  - Change the analysis, or look at a subgroup of individuals
  - Only report the analysis that happens to give $p < 0.05$
  - Called "p-hacking"

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Misuse of the p-value

- Journals: Rejecting a manuscript solely because $p > 0.05$
- Researchers: Reporting only the p-value
  - What does $p = 0.02$ "really" mean?
    - Difference is not 0
    - Nothing about the sign of the difference
    - Nothing about the magnitude of the difference
  - Really should report the estimate and a measure of its precision
- My view: p-values belong in the privacy of your office
  - Use them to avoid being misled by results that may be misleading
  - Evaluate them in conjunction in combination with everything else you know about the question and data

Discussion and disagreement about p-values

- Large literature
- Many different opinions, sometimes strident
- My sense: Often rejecting use of a p-value because it gets mis-interpreted
- Extreme situation: journal editor deciding to desk reject any manuscript with a p-value
- American Statistical Association (ASA) statement on p-values[1] with considerable supplemental material
- A more compact summary of the issues and history: Nuzzo, 2014, Nature[2]

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[1] Wasserstein, R. and Lazar, N. 2016. The ASA statement on p-values: context, process, and purpose. *The American Statistician* 70:129-133

[2] Nuzzo, R. 2014. Statistical Errors: P values, the 'gold standard' of statistical validity, are not as reliable as many scientists assume. *Nature* 506:150-152

Important Points

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- Considerable controversy about their use and interpretation
- Many ways p-values are mis-interpreted
- And many ways they are misused
    - p-hacking: choosing results because they have $p < 0.05$